

Self-Normalized Importance Sampling (SNIS)

Monte Carlo Statistical Methods

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1 Motivation

In classical importance sampling, we assume the target density $f_X(x)$ and proposal density $g(x)$ are known exactly, so the weight

$$w(x) = \frac{f_X(x)}{g(x)}$$

can be evaluated.

However, in many applications the densities are known only up to unknown normalizing constants. We may have

$$f_X(x) = \frac{\hat{f}_X(x)}{c}, \quad g(x) = \frac{\hat{g}(x)}{b},$$

where $\hat{f}_X(x)$ and $\hat{g}(x)$ are known unnormalized forms, but the constants c and b are unknown.

This prevents us from forming the classical importance weight

$$w(x) = \frac{f_X(x)}{g(x)}.$$

Self-normalized importance sampling resolves this difficulty by working entirely with unnormalized densities.

2 Setup

We want to estimate

$$p = \mathbb{E}_f[h(X)] = \int h(x) f_X(x) dx.$$

Since

$$f_X(x) = \frac{\hat{f}_X(x)}{c},$$

we write

$$p = \frac{1}{c} \int h(x) \hat{f}_X(x) dx.$$

Insert the proposal representation

$$1 = \frac{\hat{g}(x)}{\hat{g}(x)}$$

and rewrite:

$$p = \frac{1}{c} \int h(x) \hat{f}_X(x) \frac{\hat{g}(x)}{\hat{g}(x)} dx.$$

Multiply and divide by b :

$$p = \frac{b}{c} \int h(x) \frac{\hat{f}_X(x)}{\hat{g}(x)} \frac{\hat{g}(x)}{b} dx.$$

Since

$$g(x) = \frac{\hat{g}(x)}{b},$$

we obtain

$$p = \frac{b}{c} \mathbb{E}_g[h(X) \hat{w}(X)],$$

where the unnormalized weight is

$$\hat{w}(x) = \frac{\hat{f}_X(x)}{\hat{g}(x)}.$$

At this point, both b and c are unknown. To eliminate them, we write an analogous expression for the constant:

$$1 = \int f_X(x) dx = \frac{b}{c} \int \frac{\hat{f}_X(x)}{\hat{g}(x)} \frac{\hat{g}(x)}{b} dx = \frac{b}{c} \mathbb{E}_g[\hat{w}(X)].$$

Solving,

$$\frac{b}{c} = \frac{1}{\mathbb{E}_g[\hat{w}(X)]}.$$

Substitute into the formula for p :

$$p = \frac{\mathbb{E}_g[h(X) \hat{w}(X)]}{\mathbb{E}_g[\hat{w}(X)]}.$$

Thus the unknown constants cancel entirely.

3 Self-Normalized Importance Sampling Estimator

Draw

$$X_1, \dots, X_n \sim g.$$

Compute unnormalized weights:

$$\hat{w}(X_i) = \frac{\hat{f}_X(X_i)}{\hat{g}(X_i)}.$$

Normalize:

$$\bar{w}(X_i) = \frac{\hat{w}(X_i)}{\sum_{j=1}^n \hat{w}(X_j)}.$$

Define the estimator:

$$\hat{p}_n^{SNIS} = \sum_{i=1}^n \bar{w}(X_i) h(X_i).$$

This estimator is computable using only the unnormalized densities. The normalizing constants c and b never appear.

4 Interpretation

The distribution f_X is approximated by the empirical distribution

$$\sum_{i=1}^n \bar{w}(X_i) \delta_{X_i},$$

where each sample X_i contributes mass proportional to $\hat{f}_X(X_i)/\hat{g}(X_i)$. The expectation under f_X is replaced by the weighted empirical average

$$\hat{p}_n^{SNIS}.$$

The normalization ensures

$$\sum_{i=1}^n \bar{w}(X_i) = 1,$$

so the estimator behaves like an expectation under a probability distribution.

5 Properties of the SNIS Estimator

The estimator

$$\hat{p}_n^{SNIS} = \sum_{i=1}^n \bar{w}(X_i) h(X_i)$$

has the following properties:

1. It is not unbiased. The ratio of random quantities produces a bias that decreases at rate $1/n$.
2. It is consistent. Under standard assumptions,

$$\hat{p}_n^{SNIS} \rightarrow p \quad \text{almost surely.}$$

3. Its asymptotic variance can be derived using the delta method. (We do not derive it here.)
4. When the target density is known only up to a constant, SNIS is the standard and necessary estimator.

These properties make SNIS essential in Bayesian inference, filtering, and any setting where normalizing constants are unavailable.

6 Bayesian Model and Data Generation

We begin with a simple Bayesian model that produces a non-conjugate posterior, making self-normalized importance sampling (SNIS) necessary.

We observe data

$$X_1, \dots, X_n \mid \theta \sim \mathcal{N}(\theta, 1),$$

and we place a Laplace prior with scale parameter $\lambda = 1$:

$$\pi(\theta) = \frac{1}{2} e^{-|\theta|}.$$

We fix a true parameter value $\theta^* = 2$ and generate n observations:

```
true_theta = 2.0
```

```
n = 10
```

```
sigma = 1.0
```

```
samples_normal = sigma * np.random.randn(n) + true_theta
```

To understand the shape of the model components, we visualize the Laplace prior and the data-generating Gaussian distribution, along with the observed sample points.

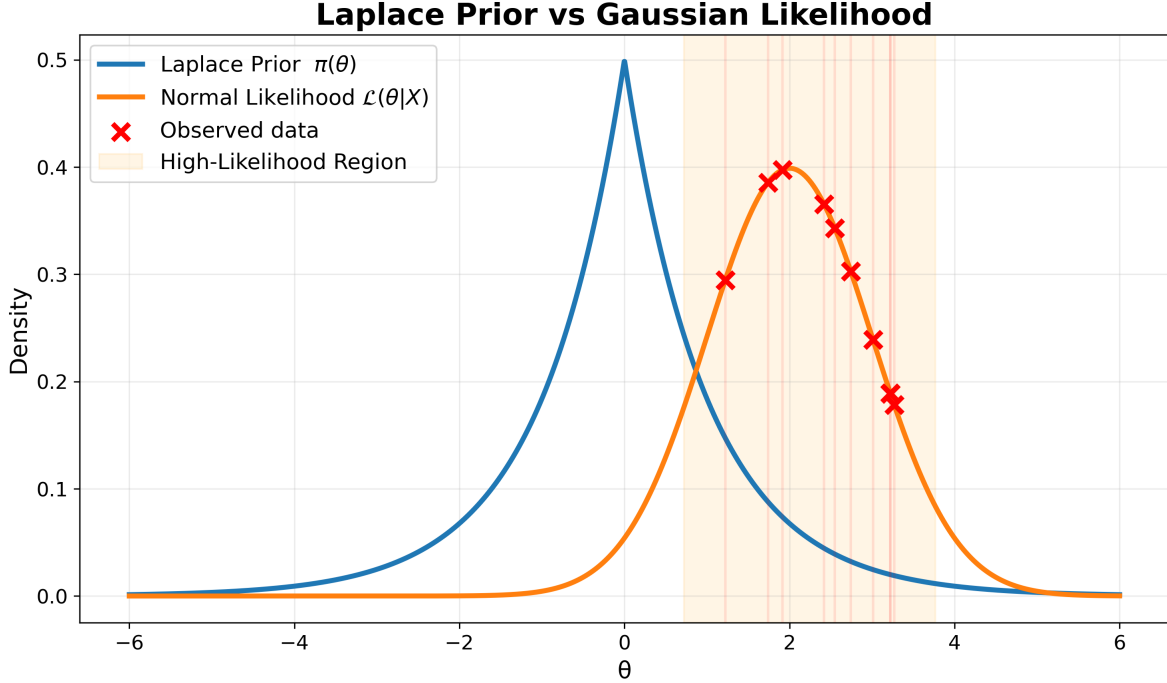


Figure 1: Laplace prior $\pi(\theta)$ and the true Gaussian data-generating distribution (viewed as a function of x). The red crosses indicate the observed samples.

7 Likelihood and Prior: Unnormalized Forms

The likelihood of θ given the full dataset is:

$$L(\theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_i - \theta)^2}{2}\right),$$

and for SNIS we keep only the θ -dependent part:

$$\hat{L}(\theta) \propto \exp\left(-\frac{1}{2} \sum_{i=1}^n (x_i - \theta)^2\right).$$

The Laplace prior is similarly taken in unnormalized form:

$$\hat{\pi}(\theta) = e^{-|\theta|}.$$

Multiplying likelihood and prior gives the unnormalized posterior:

$$\hat{f}_X(\theta) = \exp\left(-\frac{1}{2} \sum_{i=1}^n (x_i - \theta)^2 - |\theta|\right).$$

This defines our *target density* up to an unknown normalizing constant.

8 Why SNIS Is Required

The normalized posterior is:

$$\pi(\theta \mid x_1, \dots, x_n) = \frac{\hat{f}_X(\theta)}{Z}, \quad Z = \int \hat{f}_X(\theta) d\theta.$$

The quantity Z has no closed form because the prior is not conjugate to the Gaussian likelihood. Thus classical importance sampling weights

$$w(\theta) = \frac{f_X(\theta)}{g(\theta)}$$

cannot be computed.

SNIS avoids this by using only unnormalized densities.

9 Proposal Distribution

A good proposal should cover the region where the posterior has high mass. The MLE of θ is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

We choose a simple Gaussian proposal

$$g(\theta) = \mathcal{N}(\bar{x}, 1).$$

Generation and evaluation code:

```
def proposal_sampler(num_samples: int, x_data: np.ndarray) -> np.ndarray:
    theta_hat = np.mean(x_data)
    return stats.norm.rvs(loc=theta_hat, scale=1.0, size=num_samples)

def proposal_pdf(theta: np.ndarray, x_data: np.ndarray) -> np.ndarray:
    theta_hat = np.mean(x_data)
    return stats.norm.pdf(theta, loc=theta_hat, scale=1.0)
```

10 Self-Normalized Weights

We compute unnormalized weights using:

$$\hat{w}(\theta) = \frac{\hat{f}_X(\theta)}{g(\theta)},$$

where $\hat{f}_X(\theta)$ is the unnormalized posterior.

```
def posterior_unnormalized(theta, x_data):
    return np.exp(log_likelihood(theta, x_data) + log_prior(theta))

raw_weights = posterior_unnormalized(theta_samples, samples_normal) \
    / proposal_pdf(theta_samples, samples_normal)
```

Normalize the weights:

$$\bar{w}_i = \frac{\hat{w}_i}{\sum_j \hat{w}_j}.$$

```
norm_weights = raw_weights / np.sum(raw_weights)
```

11 SNIS Estimator of the Posterior Mean

For a general function $h(\theta)$, SNIS approximates

$$\mathbb{E}_f[h(\theta)] \approx \sum_{i=1}^N \bar{w}_i h(\theta_i).$$

In our example we take

$$h(\theta) = \theta,$$

so the SNIS estimator of the posterior mean is

$$\hat{\mathbb{E}}[\theta \mid x_1, \dots, x_n] = \sum_{i=1}^N \bar{w}_i \theta_i.$$

```
h_function = lambda x: x

mean_is = importance_sampling(
    num_samples,
    prop_sampler_partial,
    prop_pdf_partial,
    target_pdf_partial,
    h_function,
    snis=True
)

snis_mean = mean_is.estimate_snis
print(f"SNIS posterior mean = {snis_mean:.6f}")
```

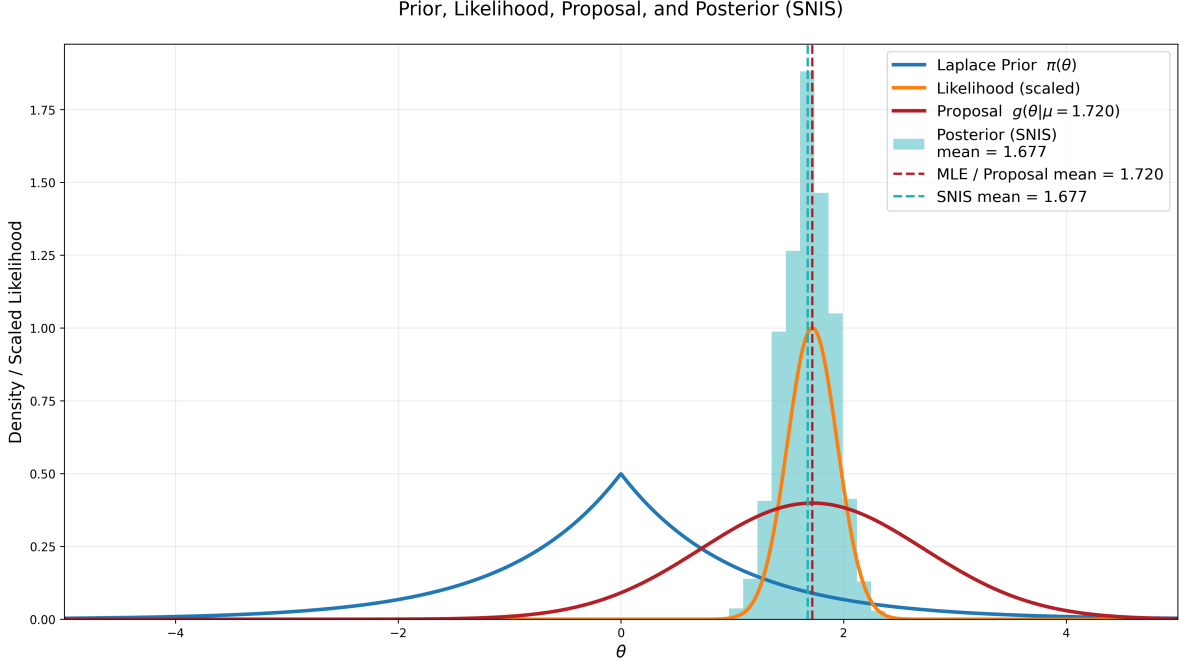


Figure 2: Laplace prior, scaled likelihood, Gaussian proposal, and posterior approximation via SNIS. Vertical lines show the proposal mean $\bar{x}$ and the SNIS posterior mean estimate.

12 Interpretation

Self-normalized importance sampling reconstructs the posterior by

$$\sum_{i=1}^N \bar{w}_i \delta_{\theta_i},$$

a weighted empirical measure whose normalized weights sum to one. Only unnormalized densities are needed; the unknown normalizing constant of the true posterior cancels automatically.

The proposal $g(\theta)$ draws samples near the likelihood peak $\bar{x}$, while the Laplace prior pulls mass toward zero. Through the weights, SNIS naturally balances these influences when computing posterior expectations.

This gives a clean, pointwise-evaluable, non-conjugate posterior for which SNIS is both necessary and effective.

13 Weight Diagnostics: Evaluating Proposal Quality and Posterior Shape

After computing the SNIS estimator, we examine the normalized and raw weights to evaluate how well the proposal overlaps with the target.

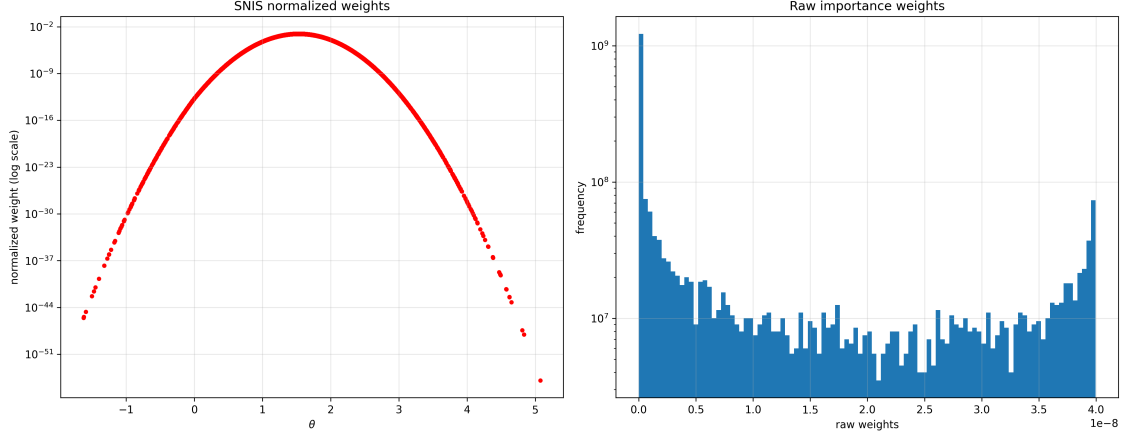


Figure 3: Left: normalized SNIS weights $\bar{w}_i$ as a function of θ_i (log scale). Right: histogram of raw weights $\hat{w}_i = \hat{f}_X(\theta_i)/g(\theta_i)$.

The normalized weights form a smooth arch: large near the posterior mode and rapidly decaying in the tails. This arch essentially *reveals the shape of the posterior itself*.

The raw-weight histogram shows many small weights (samples in low-posterior regions) and a spread of moderate weights near the peak, with no dominating outlier. This indicates the proposal $g(\theta) = \mathcal{N}(\bar{x}, 1)$ overlaps well with the target posterior and avoids weight collapse.

These diagnostics confirm that the SNIS estimator is stable and efficient for this Bayesian model.

14 Effective Sample Size (ESS)

A standard diagnostic for importance sampling is the *effective sample size* (ESS), defined for normalized weights by

$$\text{ESS} = \frac{1}{\sum_{i=1}^N \bar{w}_i^2}, \quad \bar{w}_i = \frac{\hat{w}_i}{\sum_{j=1}^N \hat{w}_j}.$$

ESS quantifies how many of the N proposal samples act as if they were effectively drawn from the target posterior. If all weights are equal, $\text{ESS} = N$; if one weight dominates, $\text{ESS} \approx 1$.

In our experiment with $N = 5000$ samples, we obtain

$$\text{ESS} \approx 1595 \quad (\text{about } 32\% \text{ of the sample}).$$

This indicates that the proposal $g(\theta) = \mathcal{N}(\bar{x}, 1)$ overlaps well with the posterior: the weights are neither uniform nor collapsed. Roughly one-third of the samples contribute meaningfully to the SNIS estimator, confirming that the approximation is stable and efficient.